HAND-OBJECT DETECTION WITH RECONSTRUCTION

Xiaohu Guo | UTD-7 | CAST
Neil Song, Alena Zhang, Kevin Lu, Veronica Fang, Aaron Liu, Jonathan He, Luke Cheng

Problem/Solution

TABLE OF CONTENTS

01 DATA COLLECTION & AUGMENTATION

02 HAND OBJECT DETECTION

03 HAND MESH RECONSTRUCTION

TERMS

Detection: Classification + Localization

- Bounding Box

Monocular RGB: Standard Color Images

Mesh Reconstruction: 3D Representation

DATA

Collection & Augmentation

01

WORKFLOW

Collect

Capture frames with cameras

Label

Bounding Box + State Classification

Augment

PASCAL_VOC

Split

Training + Validation + Testing

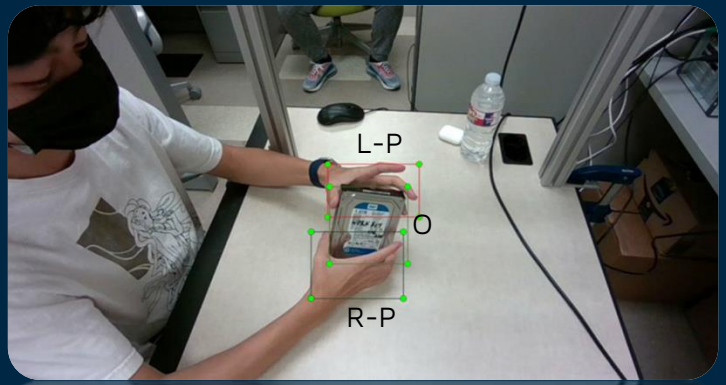
COLLECT/LABEL

Raw Photo

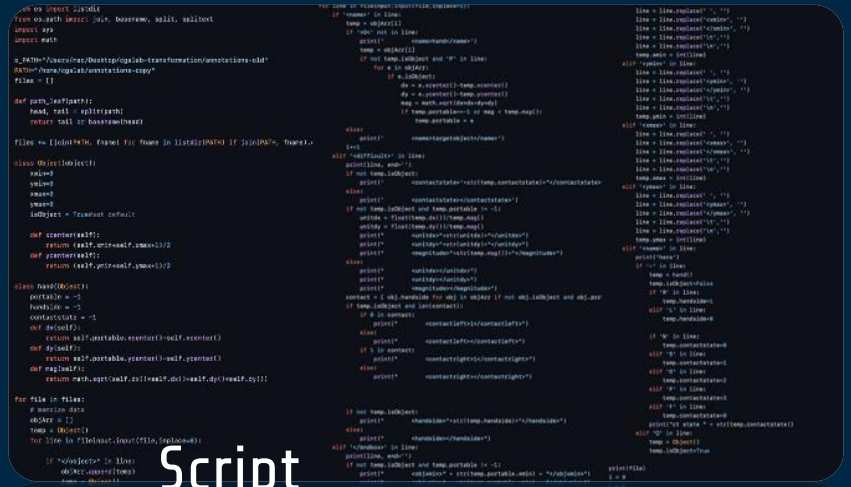
We collected over 400 photos similar to this photo. Each has at least one hand and most have an interacting object.

Annotated Photo

Each hand has a bounding box. Each box is labeled with left hand, right hand, object, etc.



AUGMENTATION



Script

The python program that converts from one format to the other


```
<annotation>
   <folder>tmp</folder>
   <filename>frame0617.jpg</filename>
   <path>/home/cgalab/tmp/frame0617.jpg</path>
   <source>
      <database>Unknown</database>
   </source>
   <size>
      <width>1280</width>
      <height>720</height>
      <depth>3</depth>
   </size>
   <segmented>0</segmented>
   <object>
      <name>R-P</name>
      <pose>Unspecified</pose>
      <truncated>1</truncated>
      <difficult>0</difficult>
      <bndbox>
         <xmin>302</xmin>
         <ymin>1</ymin>
         <xmax>663</xmax>
         <ymax>170</ymax>
      </bndbox>
   </object>
```


Before

The format in our collected dataset


```
<annotation>
   <folder>tmp</folder>
   <filename>frame0617.jpg</filename>
   <path>/home/cgalab/tmp/frame0617.jpg</path>
   <source>
      <database>Unknown</database>
   </source>
   <size>
      <width>1280</width>
      <height>720</height>
      <depth>3</depth>
   </size>
   <segmented>0</segmented>
   <object>
      <name>hand</name>
      <pose>Unspecified</pose>
      <truncated>1</truncated>
      <difficult>0</difficult>
        <contactstate>3</contactstate>
        <unitdx>0.1764333874056405</unitdx>
        <unitdy>0.9843125823683101</unitdy>
        <magnitude>53.84468404587402</magnitude>
        <handside>1</handside>
      <bndbox>
         <xmin>302</xmin>
         <ymin>1</ymin>
         <xmax>663</xmax>
         <ymax>170</ymax>
      </bndbox>
        <objxmin>406</objxmin>
        <objymin>1</objymin>
        <objxmax>578</objxmax>
        <objymax>276</objymax>
   </object>
```


After

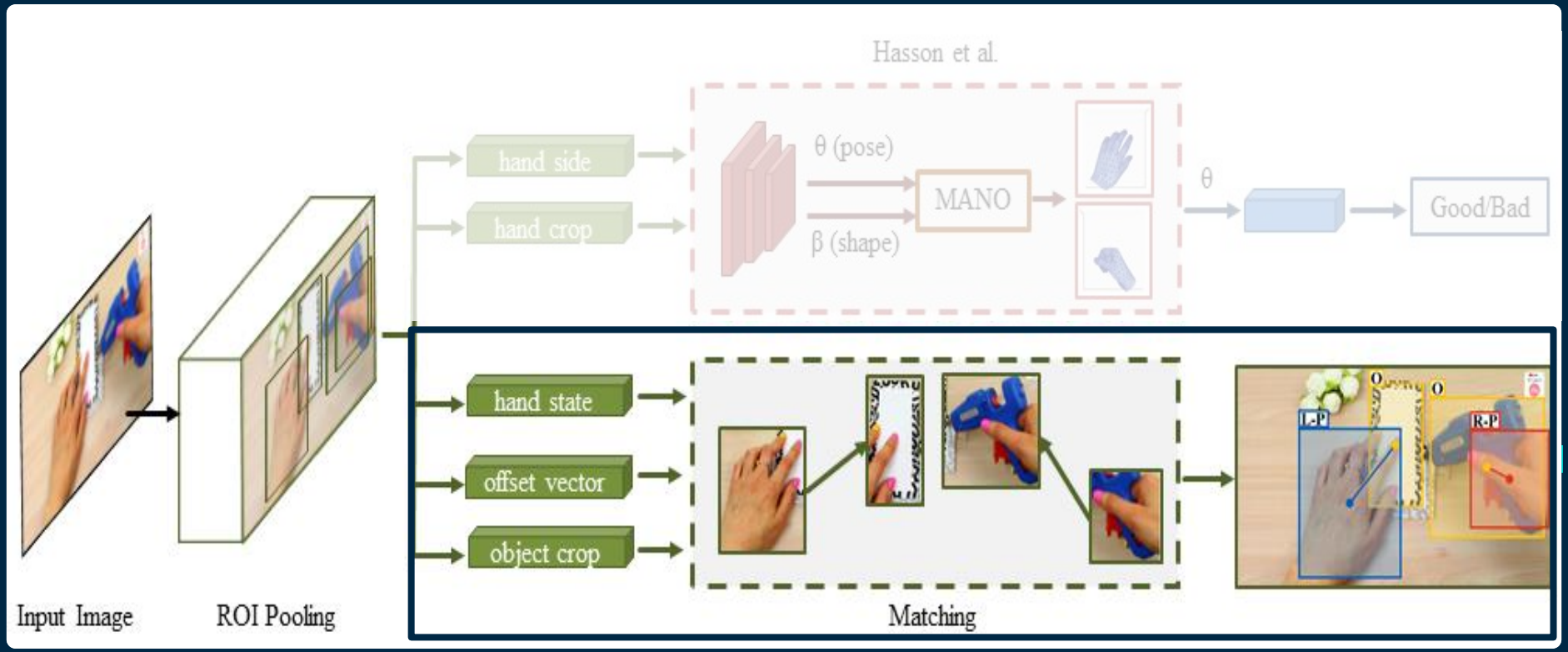
The format in the existing dataset

Detection

Hand & Object

02

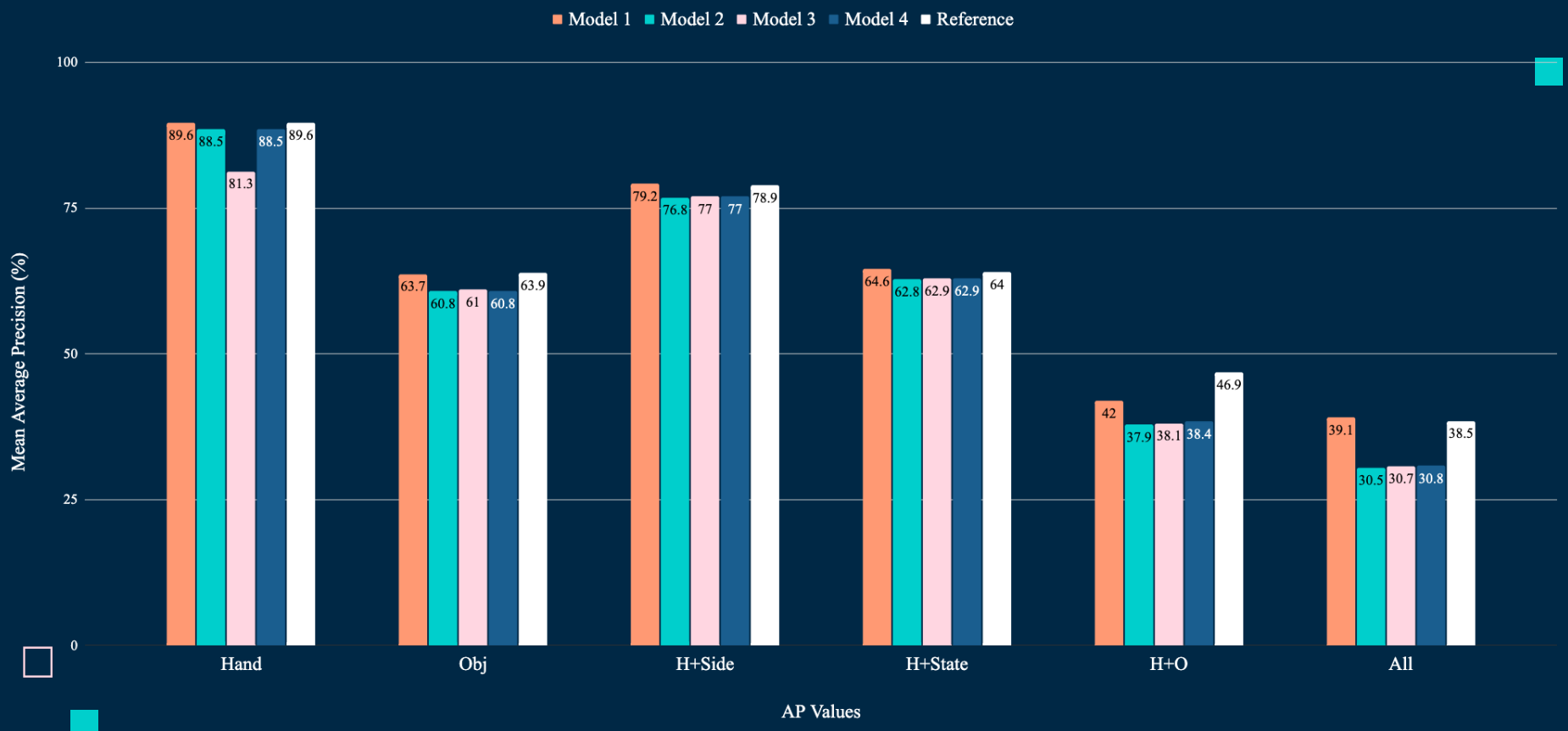
MODEL ARCHITECTURE



MODEL HYPERPARAMETERS

Model Number	Learning Rate	Epochs
1	1e-3	10
2	1e-4	8
3	1e-4	10
4	1e-5	10

MODEL PERFORMANCE



Reconstruction

Hand Mesh

03

PIPELINE

Centered crop of hand with predicted L/R state

$Crop, Side$

DETECTION

HAND ENCODER

$\Phi_{Hand} \rightarrow Pose\ (\theta), Shape\ (\beta)$

Encoded | Regressed

Differential Rendering Layer

V_{Hand}, J

MANO

MESH

Final Output

3D Model | Vertices + Faces

PIPELINE – REVERSED

3D Model | Vertices
+ Faces

Final Output

MESH

MANO

V_{Hand}, J

Differential Rendering
Layer

Encoded | Regressed

$\Phi_{Hand} \rightarrow Pose\ (\theta), Shape\ (\beta)$

HAND
ENCODER

DETECTION

$Crop, Side$

Centered crop of hand with
predicted L/R state

MANO

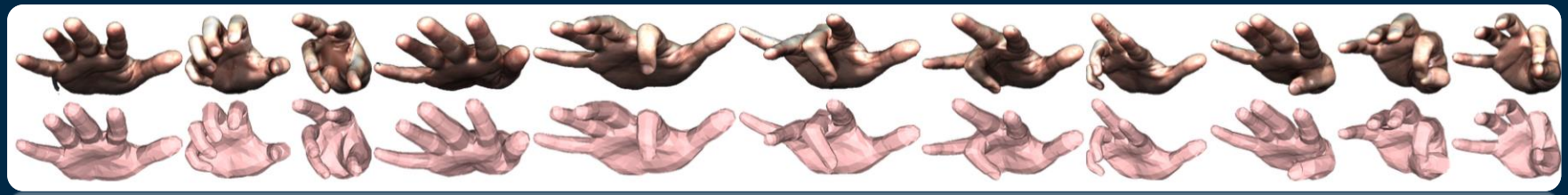
Statistical Model of the Hand

Differentiable Rendering Layer

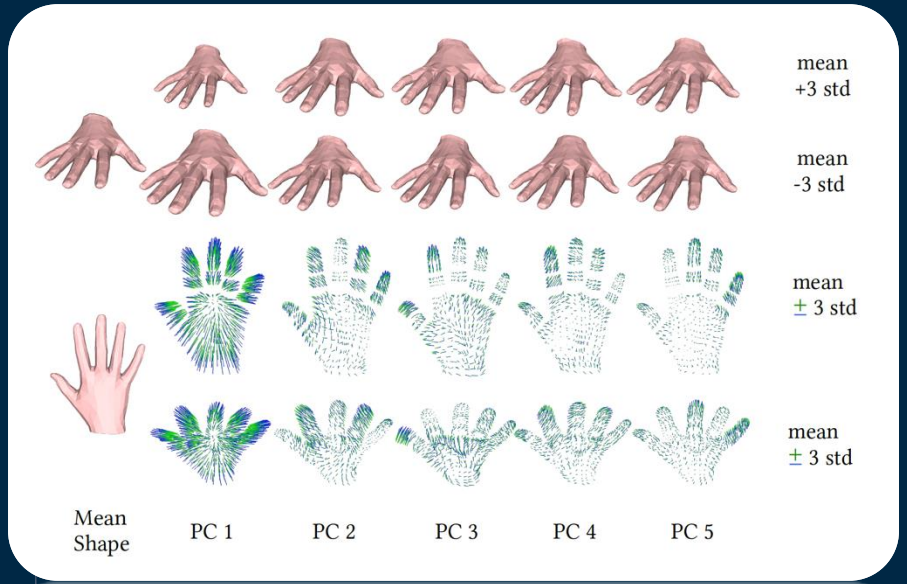
Parameters

$| \, Pose \ (\theta), Shape \ (\beta) \, |$

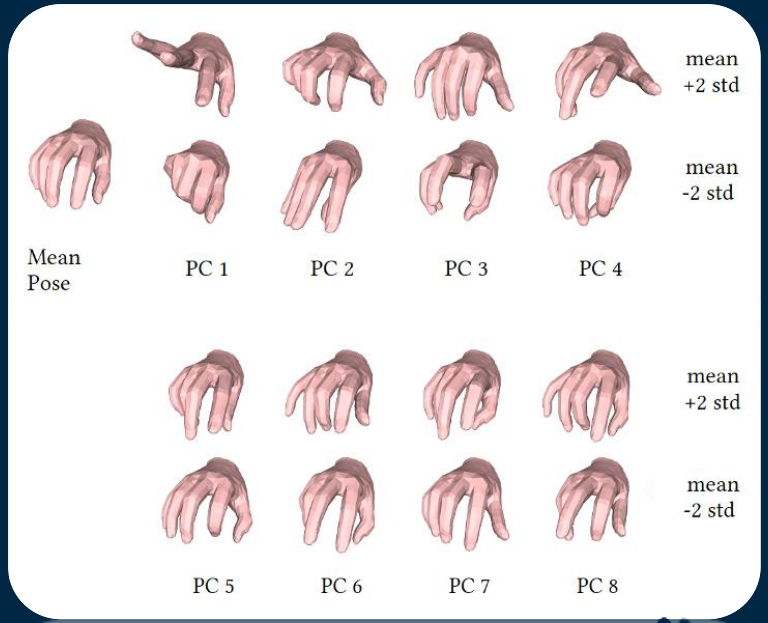
Extension of SMPL



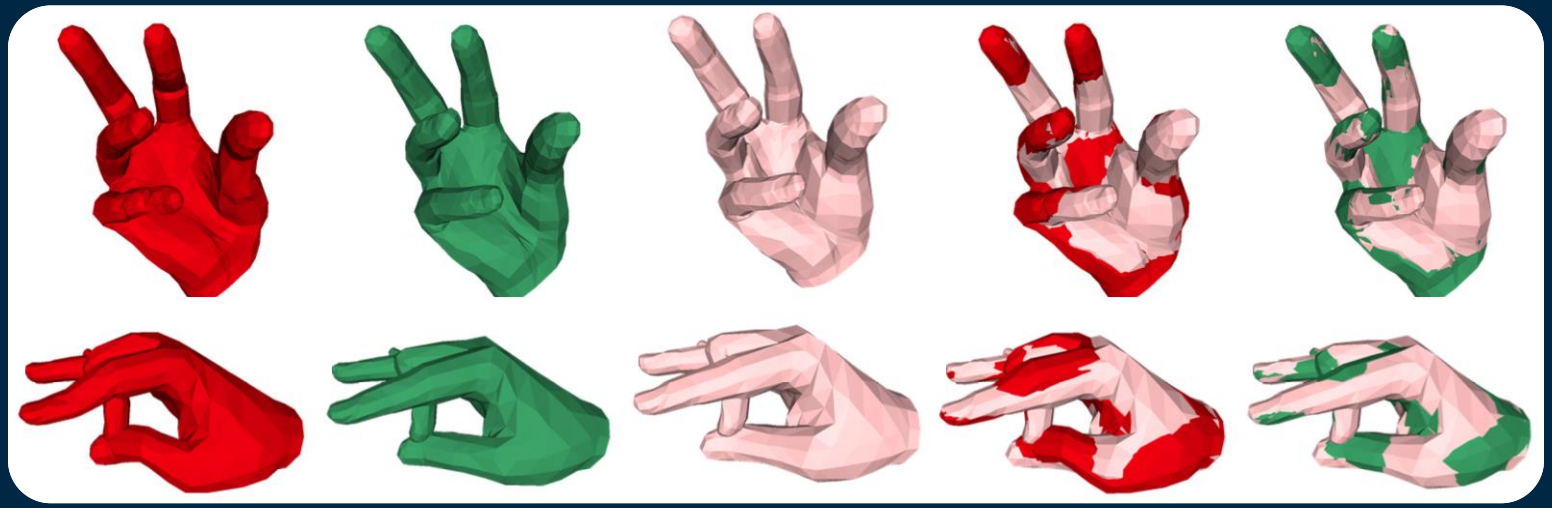
PCA SHAPE SPACE



PCA POSE SPACE



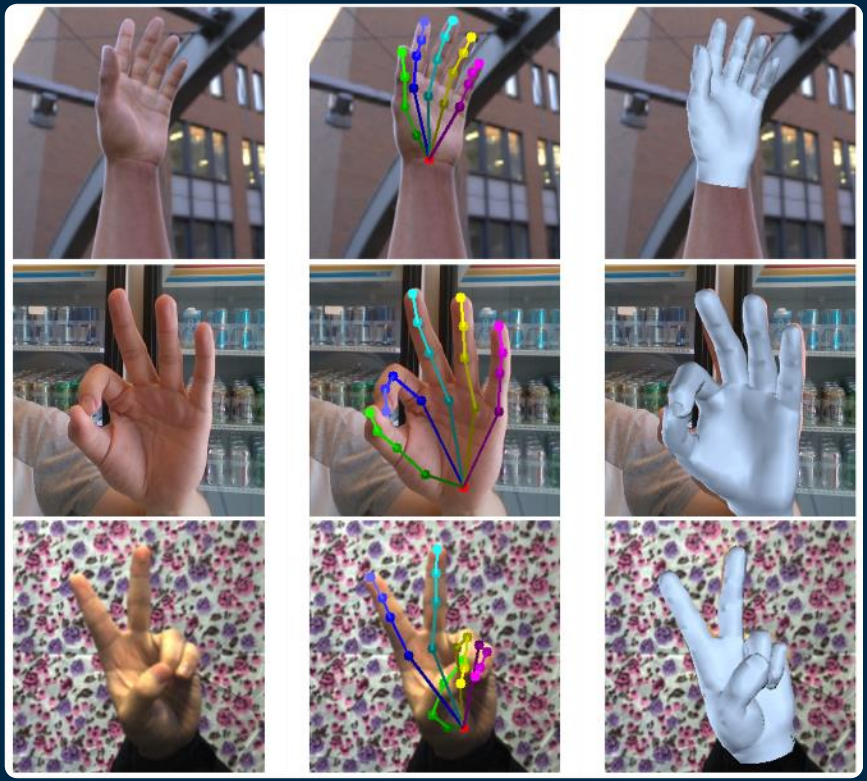
SHAPE + **POSE** BLENDSHAPES



Shape Blendshapes | Shape + Pose Blendshapes | Mesh | Shape Blendshapes Overlay | Pose Blendshapes Overlay

Hand Encoder

- ResNet-18 (ImageNet)
- Parameter
  - RGB Image
- Regression
  - Pose <- Skeleton/Joints
  - Shape <- Skin + Tissue



LOSS

Regularized - $\| \beta \|^2$

$$\mathcal{L}_{Hand} = \mathcal{L}_{\beta} + \mathcal{L}_{V_{Hand}} + \mathcal{L}_{J}$$

H_{max} to 1

Concurrency using Ray

CONCURRENCY - RAY

Reconstruction Inference

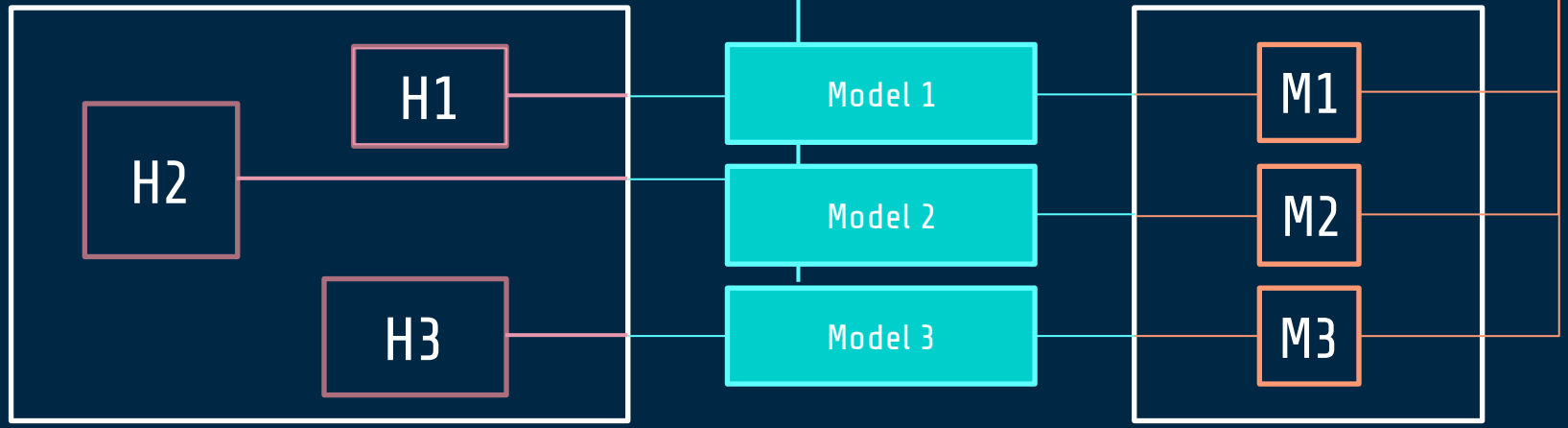
One to many hand detection mesh inference

Multiple inference as single time cost

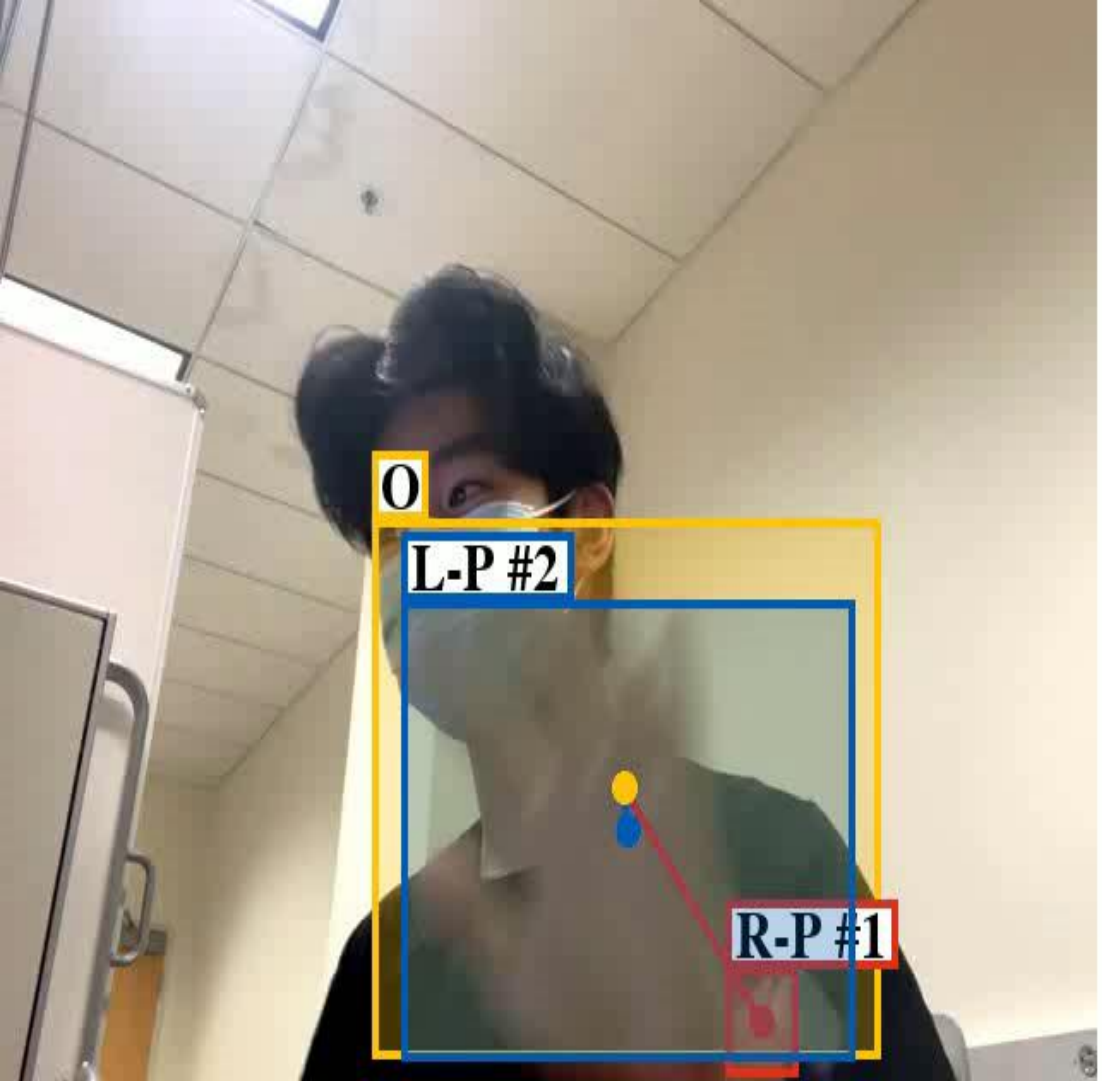
Plotting Mesh

Single-Threaded -> Multiprocessing

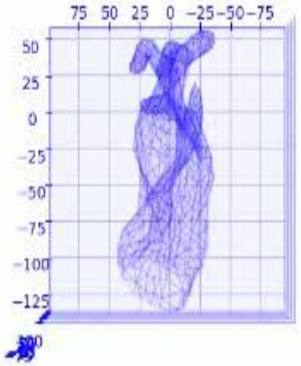
Most expensive operation (~0.2s each)



DEMO



Right #1



Left #2

CITATIONS

[1] Shan, Dandan. "100 Days of Hands." 100DOH, fouheylab.eecs.umich.edu/~dandans/projects/100DOH/download.html.

[2] Romero, Javier, et al. "Embodied Hands: Modeling and Capturing Hands and Bodies Together." Transactions on Graphics, (Proc. SIGGRAPH Asia), vol. 36, Nov. 2017, pp. 1–17. 245:1--245:17, DOI:10.1145/3130800.3130883.

[3] Lin, Darren. "LabelImg." *Github*, 27 June 2021, github.com/tzutalin/labelImg.

[4] Hui, Jonathan. "MAP (Mean Average Precision) for Object Detection." Medium, 3 Apr. 2019, jonathan-hui.medium.com/map-mean-average-precision-for-object-detection-45c121a31173.

CITATIONS

[5] Shan, Dandan, et al. "Understanding Human Hands in Contact at Internet Scale." Arxiv, University of Michigan, Johns Hopkins University, 11 June 2020, arxiv.org/pdf/2006.06669.pdf.

[6] Hasson, Yana, et al. "Learning Joint Reconstruction of Hands and Manipulated Objects." Arxiv, Inria, Departement d'Informatique De l'ENS, CNRS, PSL Research University, MPI for Intelligent Systems, Univ. Grenoble Alpes, Grenoble INP, LJK, 11 Apr. 2019, arxiv.org/pdf/1904.05767.pdf.

WEBSITE